

OSHIN REX

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EDUCATION

Cornell University

Ithaca, NY

B.S. Computer Science and Operations Research, (GPA: 3.79/4.0)

Aug 2024 – **Expected May 2028**

- **Coursework:** Algorithms, OOP, Functional Programming, Linear Algebra, Probability, Optimization, Databases, ML

EXPERIENCE

Software Engineering Intern

May – July 2026

IBM Consulting

Chicago, IL

- Architected a vendor management portal from scratch serving **60,000+ consumers** and **6,000+ vendors** using JavaScript, HTML/CSS, Dataverse, and Liquid templates, enabling vendors to view assigned purchase-of-service requests.
- Developed a **Microsoft Copilot-based intake workflow** that analyzed unstructured client service requests, extracted key requirements, and recommended the vendor partners best suited to meet each request's needs.
- Built **FetchXML queries and API integrations** across Dataverse entities to retrieve, filter, and relate vendor, purchase-of-service, and request records, powering data retrieval throughout the application's backend.
- Won **1st place of 30+ teams** in the IBM Consulting Intern Challenge by engineering a full-stack fan intelligence platform for the Seattle SuperSonics with vector embeddings, propensity scoring, and real-time streaming chat APIs.

Software Developer

Sep. 2025 – Present

Hack4Impact Cornell

Ithaca, NY

- Built and deployed a cross-platform **React Native** application for Alaska Children's Trust, digitizing 75 safety cards across 15 decks with **role-based access**, offline persistence, filtering, and user progress tracking.
- Built and deployed a full-stack inventory platform using **React, Express.js, Firestore, and REST APIs**, replacing manual spreadsheet-based tracking with a centralized database of inventory, sales, and location data.
- Designed separate **authenticated staff and public read-only workflows** with search and filtering, enabling staff to make warehouse updates while allowing consumers to check inventory without creating accounts.

Software Developer Intern

Dec. 2025 – Feb. 2026

GreenSand AI

Remote

- Engineered an automated **Django** pipeline processing **5,000+ PDFs**, converting scanned documents into searchable, structured text with word-level coordinates to power interactive in-document text highlighting.
- Cut end-to-end processing time by **30%** by integrating MinerU to improve parsing and clean scanned-text output.

PROJECTS

High-Performance Limit Order Book | C++20, STL, CMake

May – June 2026

- Processed **1M+ synthetic orders** with deterministic FIFO execution and **429 ns median matching latency** by implementing a price-time priority engine supporting market orders, limit orders, partial fills, and cancellations.
- Reduced order cancellation latency to **1.38 μ s**, achieving a **99.4% improvement** over a naive linear-scan baseline by indexing orders by ID with hash maps and organizing price levels with ordered data structures.
- Achieved **1.14M orders/sec** throughput at sub-**3 μ s p99 latency** (matching: 1.9 μ s, cancellation: 3.0 μ s) across 1M+ randomized orders by developing a custom C++20 benchmarking harness to measure matching, insertion, and cancellation performance (median and p99 latency) under sustained, high-throughput synthetic order flow.
- Reduced memory usage by **12.5%** and heap allocations by **26.9%** by profiling hot execution paths and optimizing object lifetimes, memory access patterns, and cache locality throughout the core order matching engine.

Case Match (Agentic Memory) | Python, LangGraph, CockroachDB, AWS Bedrock, Docker

July – Aug. 2026

- Evaluated semantic retrieval across **48+ consulting engagements** using precision and recall at 5 results, benchmarking embedding-based search against service-line similarity as a relevance proxy across representative consulting queries.
- Built a self-correcting **LangGraph retrieval workflow** using LLM-based relevance evaluation across business problems, solution approaches, and outcomes, expanding from 5 to 8 candidates across industries when initial results are insufficient.
- Achieved **175 ms median vector-search** and **330 ms end-to-end latency** across 48 queries using 1,024-dimensional AWS Titan text embeddings stored and queried with **CockroachDB's native vector search**.
- Overrode the top vector-similarity match in **52% of cases** across 48-query blind evaluations, using AWS Bedrock and Pydantic to evaluate business problem, solution, and cross-industry transferability when ranking consulting precedents.

TECHNICAL SKILLS

Languages: Python, C, C++, Java, JavaScript, SQL, OCaml

Backend & Data: Django, FastAPI, Express.js, REST APIs, PostgreSQL, CockroachDB, Redis, Kafka, Firestore

Cloud & Tools: AWS (EC2, S3, Bedrock), Docker, Git, CMake, React, React Native